大学物理综合习题集

一、基础运动学与力学

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1. The resolution limit of an optical microscope is determined by ( ).

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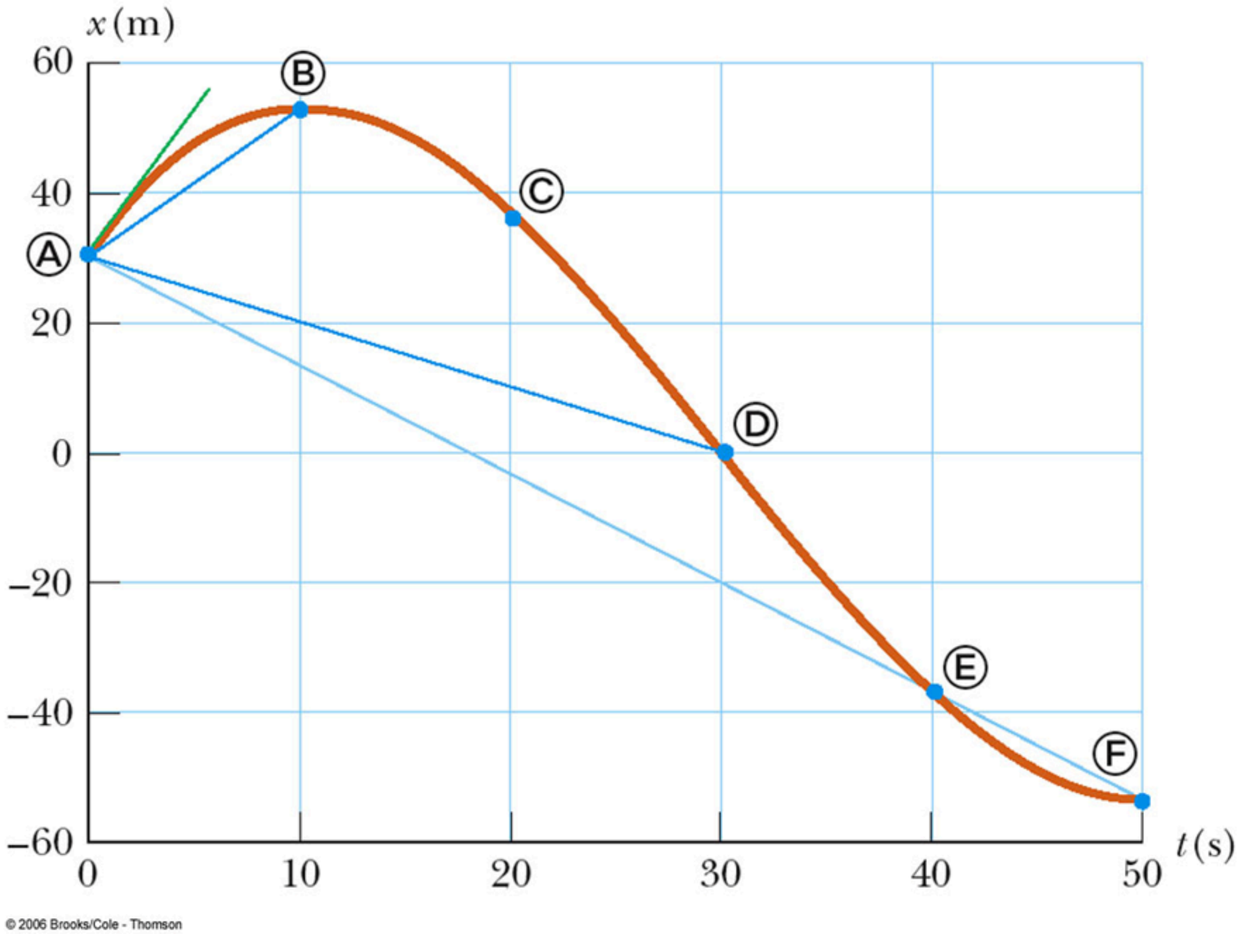
2. How to get the average velocity and instantaneous velocity in a x-t curve?



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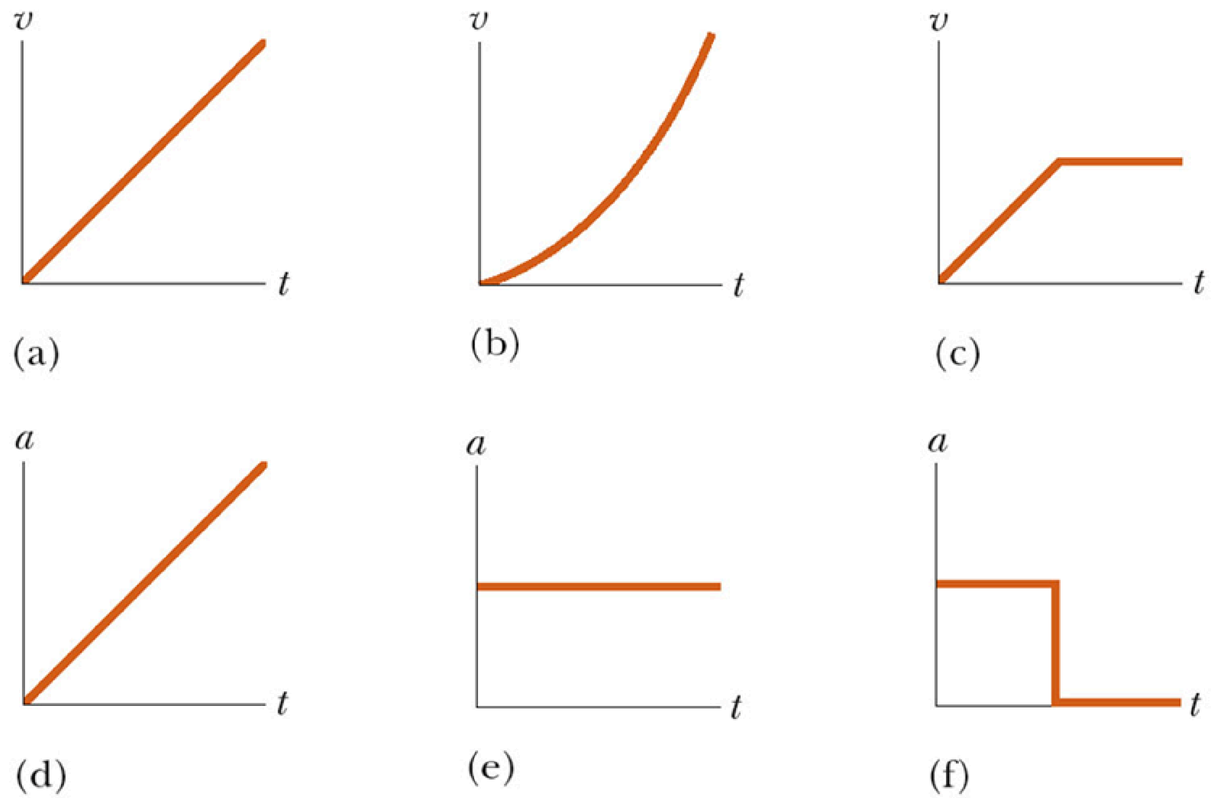
3. Match v(t) to a(t).



4. A car and a motorcycle are racing from rest. Initially the motorcycle takes the lead, but then the car blows past the motorcycle. The motorcycle first takes the lead because its (constant) acceleration $a_m = 8.40\,\mathrm{m/s^2}$ is greater than the car's (constant) acceleration $a_c = 5.60\,\mathrm{m/s^2}$, but it soon loses to the car because it reaches its greatest speed $v_m = 58.8\,\mathrm{m/s}$ before the car reaches its greatest speed $v_e = 106\,\mathrm{m/s}$. How long does the car take to reach the motorcycle?

5. Figure shows the following three vectors:
$\vec{a} = (4.2\,\mathrm{m})\vec{i} - (1.5\,\mathrm{m})\vec{j}$ $\vec{b} = (-1.6\,\mathrm{m})\vec{i} + (2.9\,\mathrm{m})\vec{j}$ $\vec{c} = (-3.7\,\mathrm{m})\vec{j}$
To add these vectors, find their net x-component and their net y-component. What is

their vector sum $\vec{r}$ which is also shown?

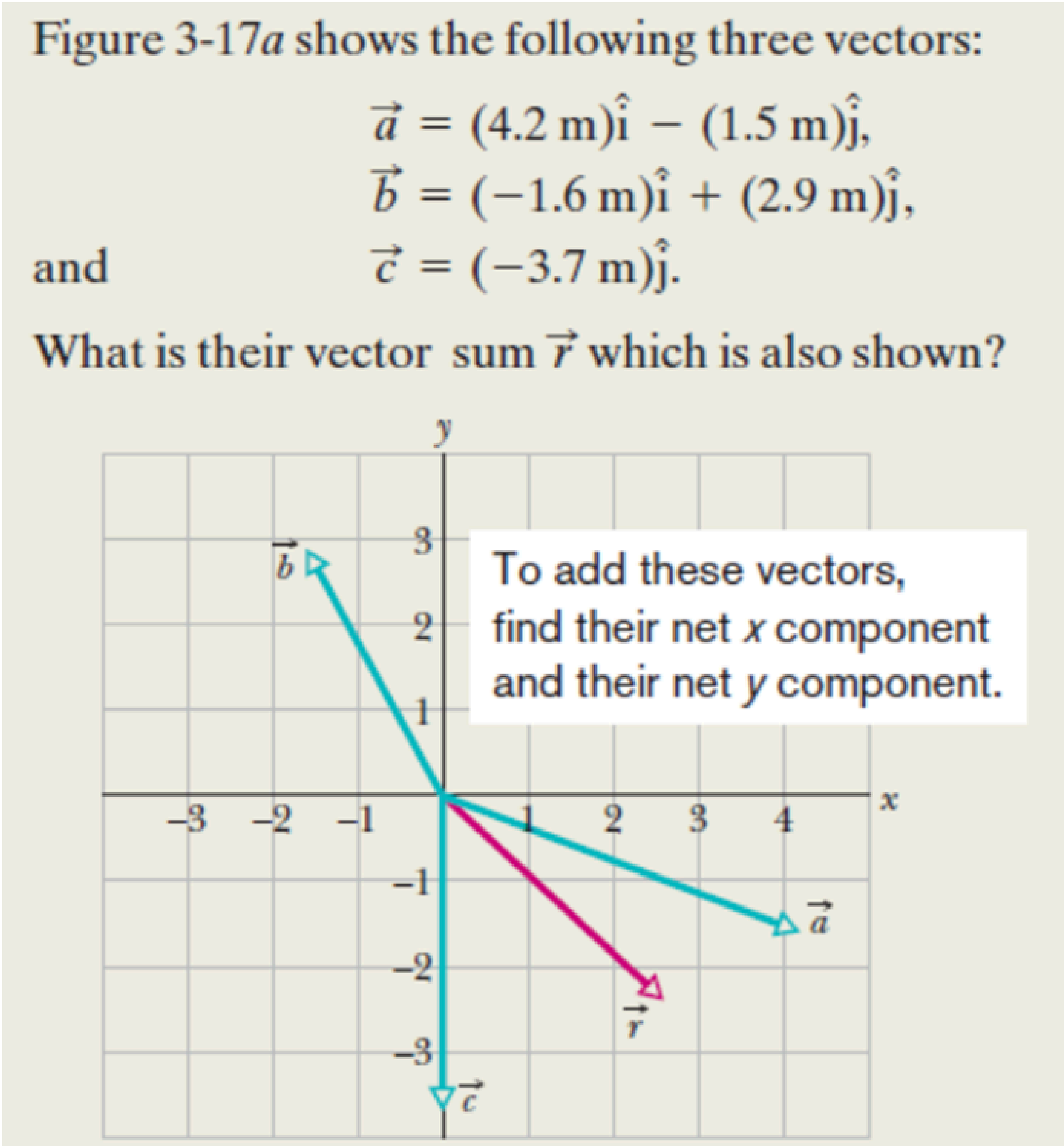


Figure 3-17*a* shows the following three vectors:

$$\vec{a} = (4.2\text{ m})\hat{\mathrm{i}} - (1.5\text{ m})\hat{\mathrm{j}},$$
$$\vec{b} = (-1.6\text{ m})\hat{\mathrm{i}} + (2.9\text{ m})\hat{\mathrm{j}},$$
and $$\vec{c} = (-3.7\text{ m})\hat{\mathrm{j}}.$$

What is their vector sum $\vec{r}$ which is also shown?

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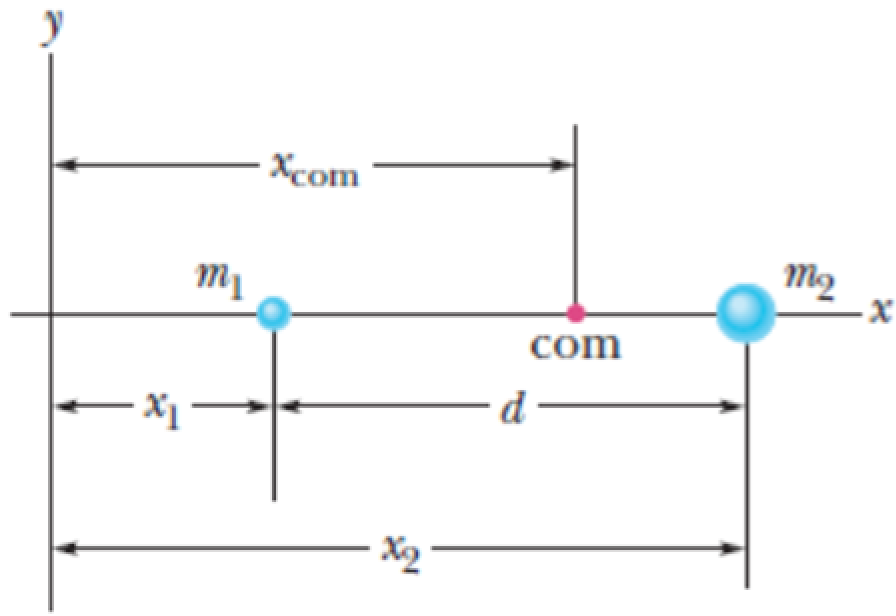
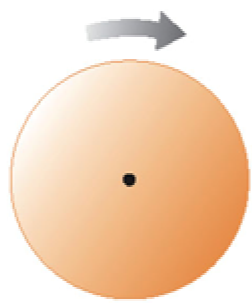
6. The center of mass of the system made up by the small ball with mass m_1 at x_1 and big ball with mass m_2 at x_2 is ( ).



- 7. The disk rotates clockwise, the direction of angular velocity is ( ).


(a)

- 8. An object of mass m moves right with speed v . It collides head-on with an object of mass $3m$ moving left with speed $v/3$. If the two stick together to have mass $4m$, what is the speed of the combined object?
  Answer: ( )
  (A) 0 (B) $v/2$ (C) v (D) $2v$

- 9. An 8.00 kg object moving east at 15.0 m/s on a frictionless horizontal surface collides with a 10.0 kg object at rest. After the collision, the 1st object moves South at 4.00 m/s.
  Question 1: Find the velocity of the 2nd object.
  Question 2: What percentage of the initial kinetic energy is lost in the collision?

- 10. A helicopter rotor turns at 320 revs/min. How fast is that in radians per second?

- 11. A race car accelerates uniformly from a speed of 40.0 m/s to 60.0 m/s in 5.00 s around a circular track of radius 400 m. When the car reaches a speed of 50.0 m/s, find the:

(1) Centripetal acceleration.
(2) Angular speed.
(3) Tangential acceleration.
(4) The magnitude of the total acceleration.

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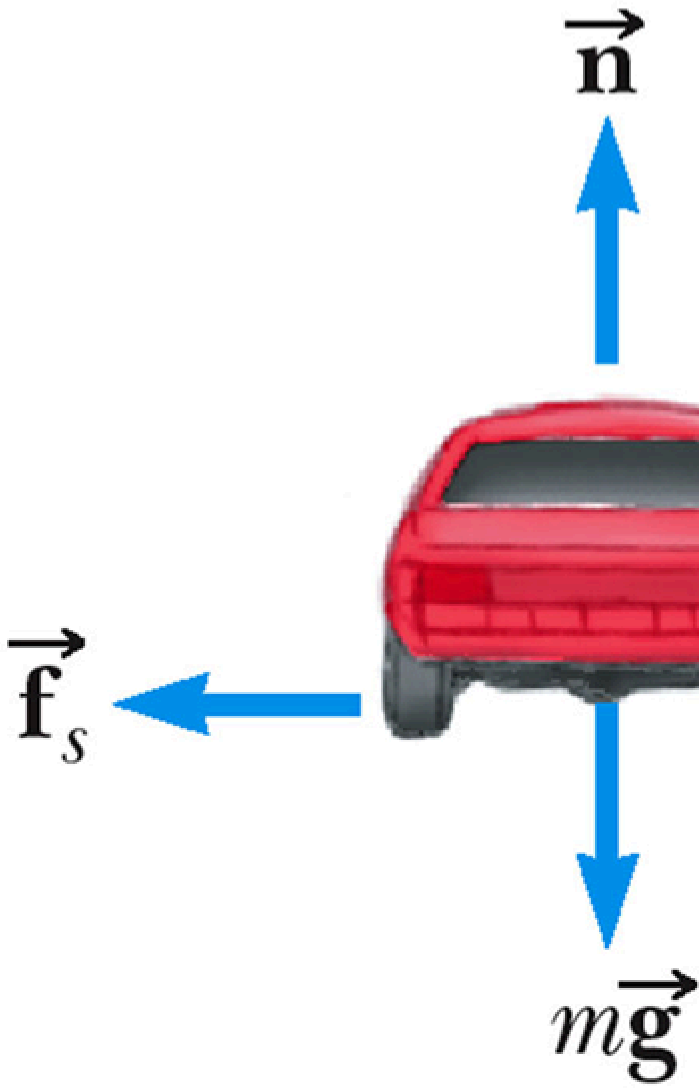
12. Friction is the force that produces the centripetal acceleration, r , μ , g are given, prove that $v = \sqrt{\mu r g}$.



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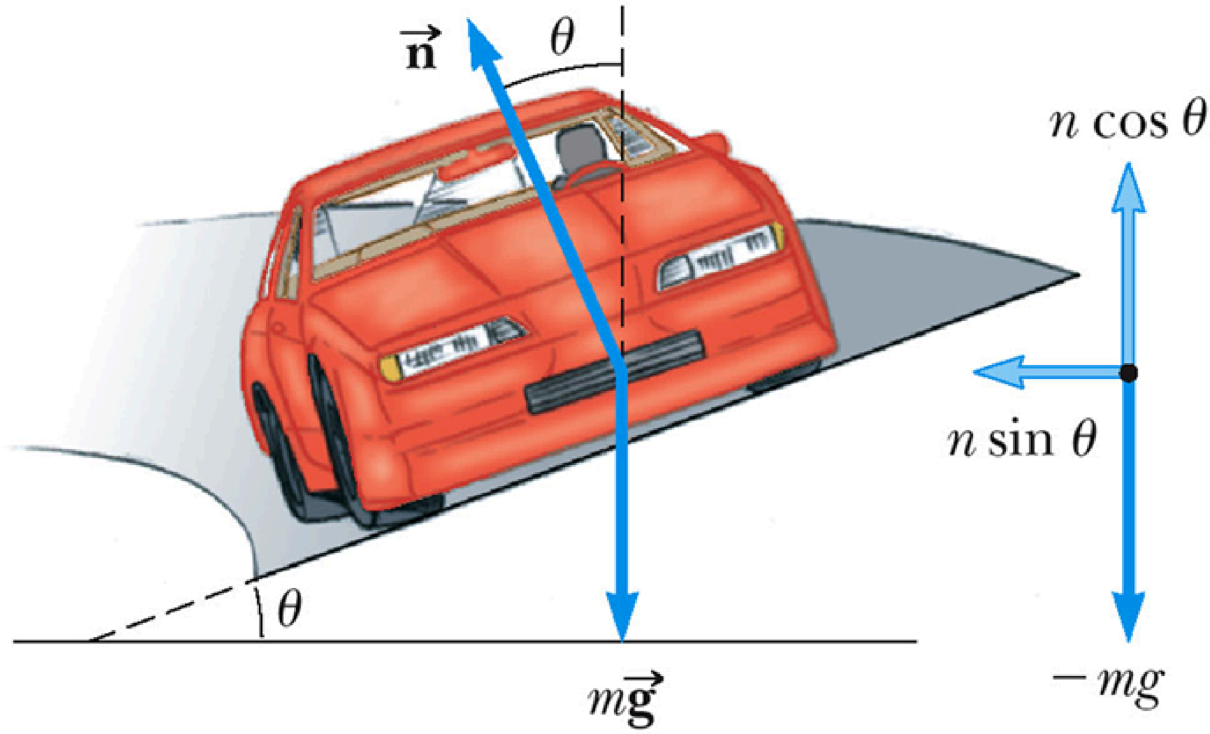
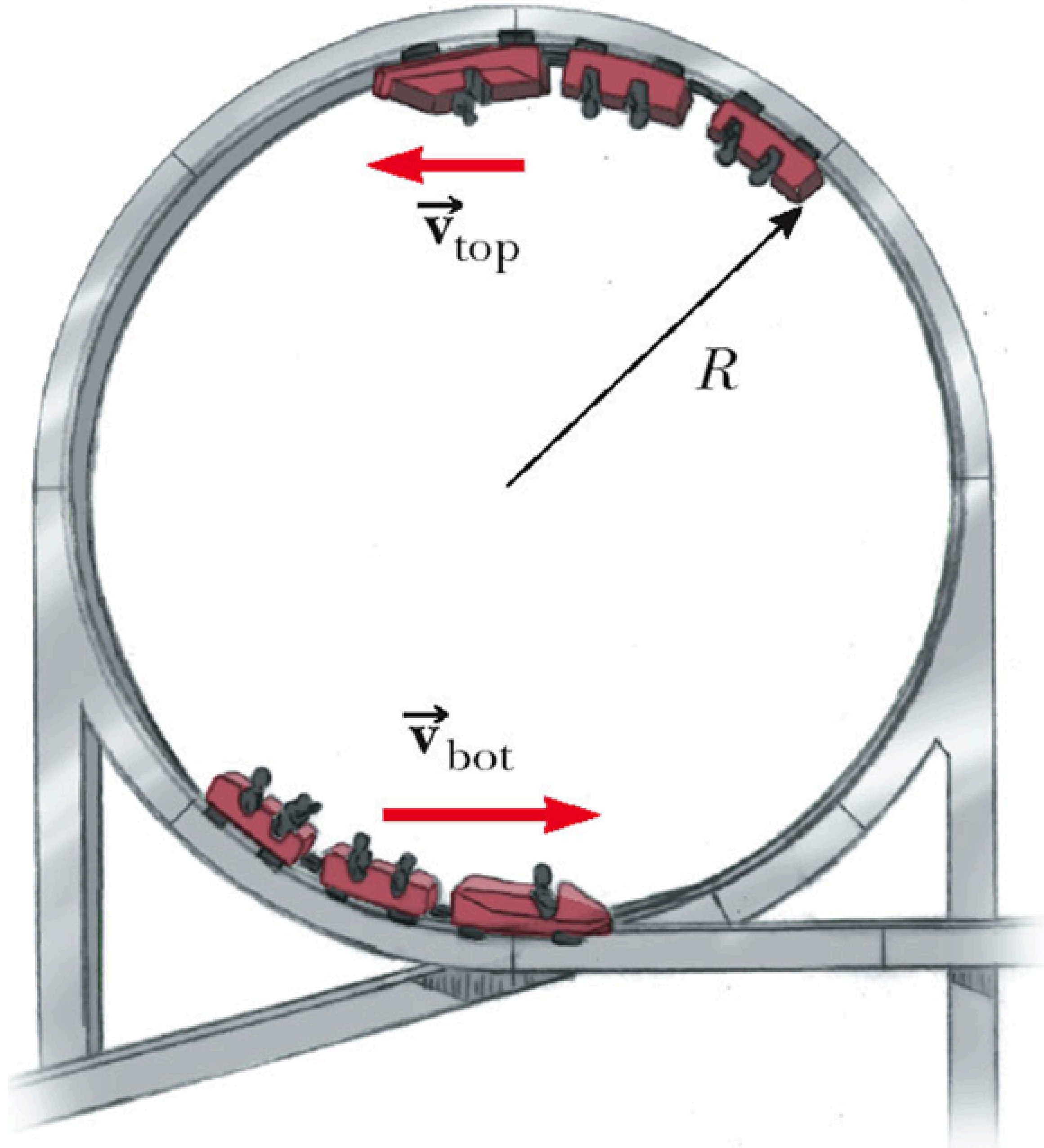
13. A component of the normal force adds to the frictional force to allow higher speeds. In this example of circular motion sustained without friction, prove that $\tan\theta = \frac{v^2}{rg}$ (or

$a_c = g \tan\theta$).



14. Look at the forces at the top of the circle. The minimum speed at the top of the circle is?



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15. How to calculate the first cosmic velocity?

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16. How to calculate the second cosmic velocity?

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17. According to Kepler's First Law, all planets move in ( ) orbits with the Sun at ( ).

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18. According to Kepler's Third Law, the square of the orbital period of any planet is proportional to ( ) of the average distance from the Sun to the planet.

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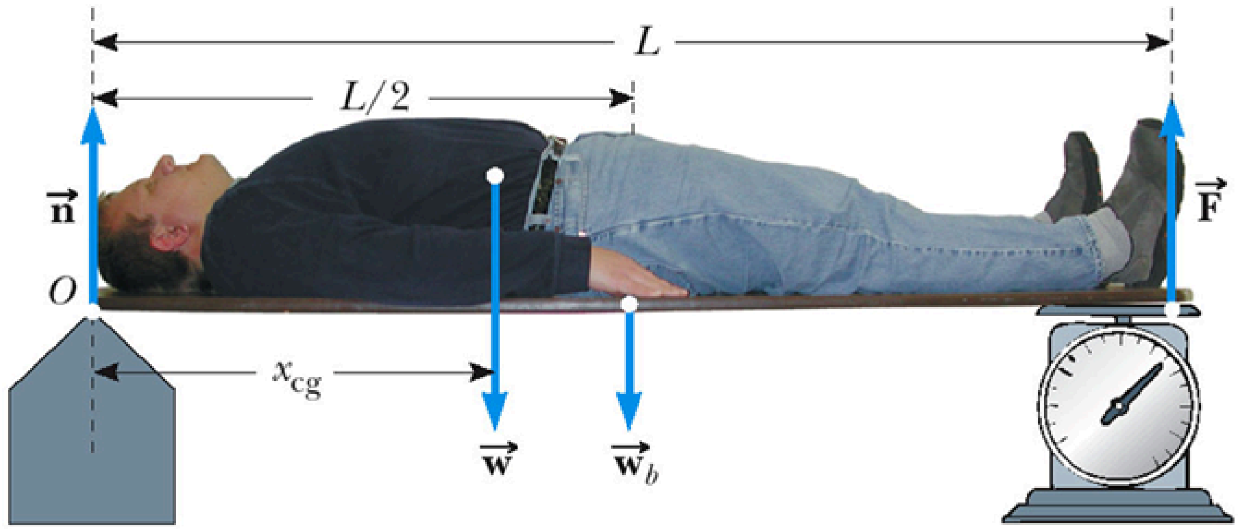
19. Consider a person with $L = 173\,\mathrm{cm}$ and weight $w = 715\,\mathrm{N}$ laying on a board with weight $w_b = 49\,\mathrm{N}$; a scale has a force reading of $F = 350\,\mathrm{N}$. Find the person's center of gravity.



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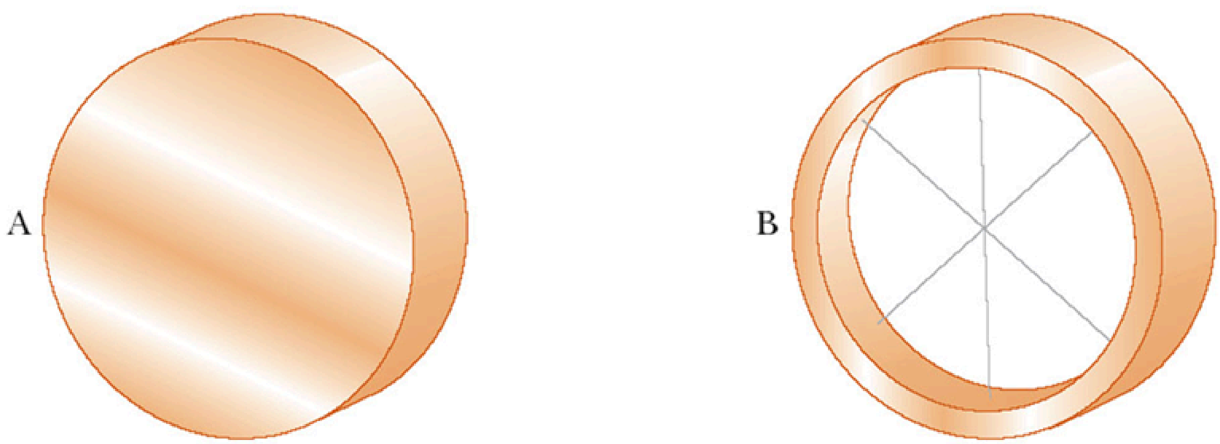
20. The two rigid objects have the same mass, radius, and angular speed. If the same braking torque is applied to each, which will take longer to stop? A or B? Why?



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21. When a skater is rotating on the surface of ice, with hands and feet drawn closer to the body, the skater's angular speed will increase or decrease? Why?

二、振动与波动

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1. Simple Harmonic Motion is motion that occurs when the net force along the direction of motion obeys ( ). The force is ( ) to the displacement and always directed toward the ( ) position.

2. Right or wrong: All periodic motion over the same path are simple harmonic motion.
3. In general, the motion of a pendulum with big angles is not simple harmonic; only when the angles are less than ( ), it becomes simple harmonic.
4. What is the difference between the motion of a simple pendulum and simple harmonic motion in terms of period?
5. In a transverse wave, each element that is disturbed moves in a ( ) direction to the wave motion.
6. In a longitudinal wave, the elements of the medium undergo displacements ( ) to the motion of the wave.
7. If two traveling waves are moving through a medium, the resulting wave is found by ( ) together the displacements of the individual waves point by point; when they pass through each other they will ( ) be altered.
8. In constructive interference, if the two waves have the same frequency and amplitude, they are called in the same ( ). The combined wave has the ( ) frequency and a ( ) amplitude.
9. What is the difference when a traveling wave is reflected from a fixed end and from a free end?

三、固体、液体与气体

1. Describe the interactions and movement of molecules in solids, liquids and gases.
2. What is the difference between crystalline and amorphous solids?
3. Explain the content of unit cell theory.
4. According to the edge ( ) of the unit cell and the ( ) between the crystal faces, crystals are divided into seven types, usually called the seven crystal systems.

5. What is the difference between strain and stress?

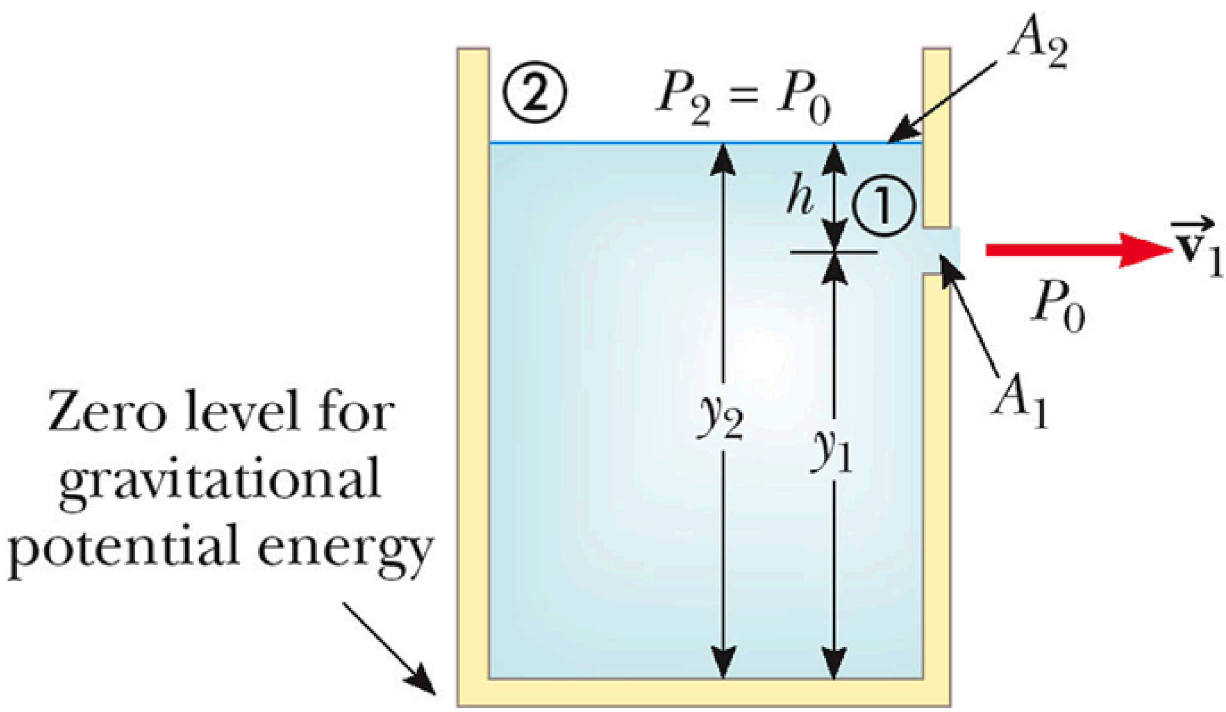
6. Draw the curve to reveal the relation of strain and stress, and label the two critical points.

7. The pressure at the bottom of a glass filled with water ($\rho = 1000\,\mathrm{kg/m^3}$) is P . The water is poured out, and the glass is filled with ethyl alcohol ($\rho = 806\,\mathrm{kg/m^3}$). Now the pressure at the bottom of the glass is.
(a) Smaller than P (b) Larger than P (c) Equal to P (d) Indeterminate

8. You buy a gold crown and want to know if it is really gold. Its weight is 7.84 N. In water, the weight is 6.86 N. Is it gold? ($\rho_{\text{gold}} = 19.3\,\mathrm{g/cm^3}$)

9. Explain the origin of the lift force in an airplane wing with Bernoulli's Equation.

10. Consider a tank with a hole. The top is open to the atmosphere. Determine the speed at which water leaves the hole if it is 0.500 m below the water level. Where does the stream hit the ground if the hole is 3.00 m above the ground?



四、热学基础

1. The exchange of energy between objects because of ( ) is called heat.

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2. What is the content of the Zeroth Law of Thermodynamics?

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3. Tell the difference between the Seebeck Effect and the Peltier Effect.

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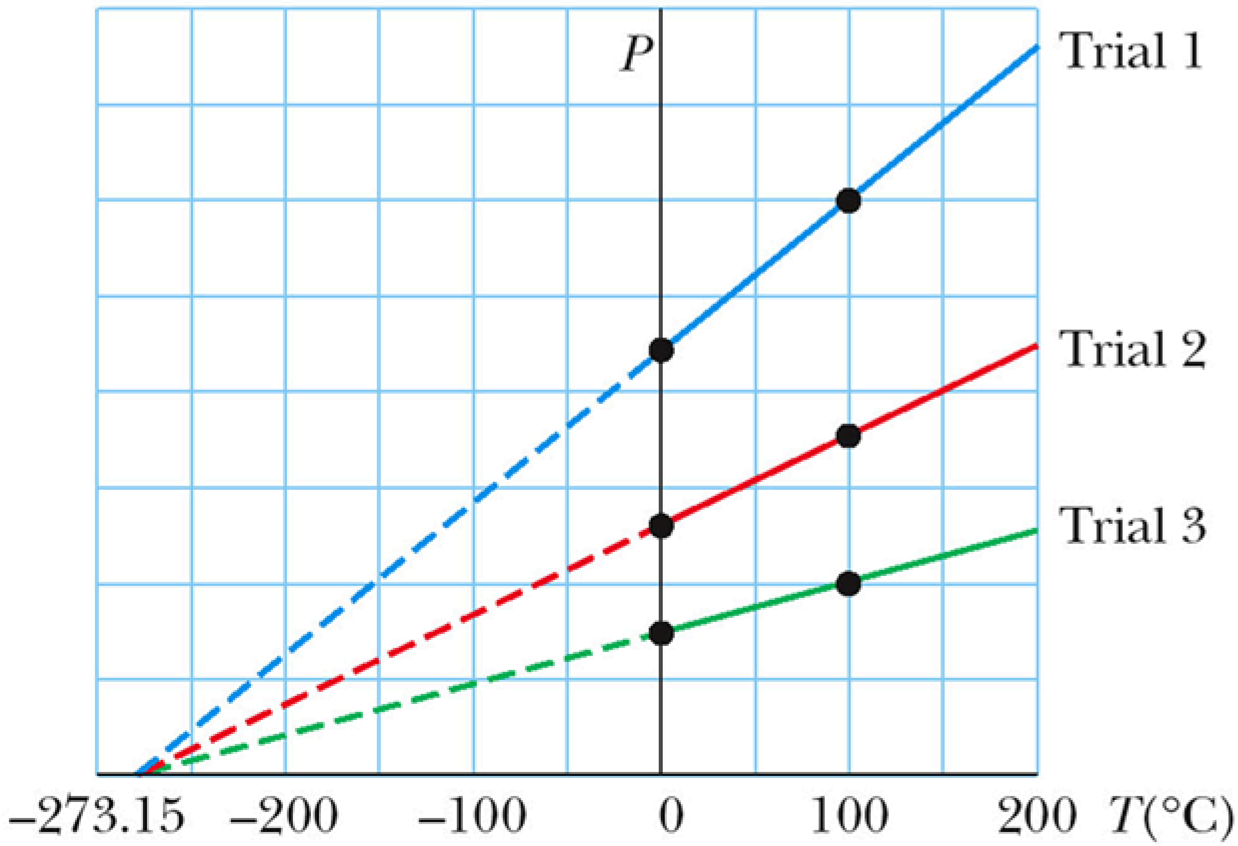
4. Explain why all gases extrapolate to the same temperature at zero pressure.



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5. Explain why absolute zero cannot be reached with the ideal gas state equation.

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6. The thermal expansion of an object is a consequence of the change in the ( ) between its constituent atoms or molecules.

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7. Explain the origin of thermal expansion.

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8. Explain the structure of graphite.

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9. Determine the increase in ocean depth due to linear expansion from an increase of 1°C owing to global warming effects. Assume a current average ocean depth of $L_0 = 4000\,\mathrm{m}$

and use $\alpha = 6.90\times10^{-5}(^{\circ}C)^{-1}$.

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10. Explain the unusual behavior of water: As the temperature of water increases from 0°C to 4°C, it contracts and its density increases. Above 4°C, water exhibits the expected expansion with increasing temperature. The maximum density of water is $1000\,\mathrm{kg/m^3}$ at 4°C.

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11. A steel strut near a ship's furnace has a mass of 1.57 kg. It absorbs thermal energy from the furnace in the amount of $2.50\times10^{5}\,\mathrm{J}$. Find its change in temperature if it has a specific heat $c = 448\,\mathrm{J/kg/^{\circ}C}$.

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12. Explain why: During the day, the wind blows from the sea surface to the land, while at night, the wind blows from the land to the sea surface.

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13. A 125 g block of unknown substance with $T = 90^{\circ}C$ is placed in a calorimeter containing 0.326 kg of water at 20°C. The system achieves an equilibrium temperature of 22.4°C. Find the specific heat c for the unknown substance. Note that the specific heat of water is $c_w = 4190\,\mathrm{J/kg/^{\circ}C}$.

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14. Phase changes involve a change in the ( ) but no change in ( ).

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15. 6 kg of ice at -5°C is added to 30 liters of water at 20°C. What is the temperature of the water when it comes to equilibrium?

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16. Explain sublimation in terms of intermolecular forces.

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17. Explain sublimation in terms of thermodynamic equilibrium.

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18. A copper slug whose mass m_c is 75 g is heated in a laboratory oven to a temperature T of 312°C. The slug is then dropped into a glass beaker containing a mass $m_w = 220\,\mathrm{g}$ of water. The heat capacity C_b of the beaker is 45 cal/K. The initial temperature T_i of the water and the beaker is 12°C. Assuming that the slug, beaker, and water are an isolated system and the water does not vaporize, find the final temperature T_f of the system at thermal equilibrium.

五、热力学定律

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1. Thermodynamics is the study of heat and its relationship to ( ), ( ) and matter.

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2. What is the nature of Heat (Q) and Work (W)?
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3. The First Law of Thermodynamics asserts that energy cannot be ( ) or ( ) in an isolated system; it can only be transformed from one ( ) to another or transferred between systems.
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4. How to understand that the First Law of Thermodynamics provides a connection between microscopic and macroscopic worlds?
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5. An ideal gas absorbs 5000 J of energy while doing 2000 J of work on the environment during a constant pressure process.
(a) Compute the change in internal energy of the gas.
(b) If the internal energy drops by 4500 J and 2000 J is expelled from the system, find the change in volume assuming a constant pressure at $1.01 \times 10^5\,\mathrm{Pa}$.
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6. Let 1.00 kg of liquid water at 100°C be converted to steam at 100°C by boiling at standard atmospheric pressure (which is 1.00 atm or $1.01 \times 10^5\,\mathrm{Pa}$). The volume of that water changes from an initial value of $1.00 \times 10^{-3}\,\mathrm{m}^3$ as a liquid to 1.671 m^3 as steam.
(a) How much work is done by the system during this process?
(b) How much energy is transferred as heat during the process?
(c) What is the change in the system's internal energy during the process?
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7. According to the Second Law of Thermodynamics, heat cannot ( ) transfer from a lower temperature body to a higher temperature body. It is ( ) to extract heat from a single source and convert it completely into useful work without causing other side effects.
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8. The Second Law of Thermodynamics is a fundamental principle that governs the ( ) of natural processes and the behavior of energy within systems.
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9. Entropy is a measure of ( ) or randomness in a system. The Second Law asserts that the total entropy of an isolated system tends to ( ) over time.
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10. Methods of Heat Transfer include ( ), ( ) and ( ).
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11. Explain the reason causing conduction of heat, and why metals are good conductors?
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12. Figure shows the cross section of a wall made of white pine of thickness L_a and brick of thickness $L_d(= 2.0L_a)$, sandwiching two layers of unknown material with identical thicknesses and thermal conductivities. The thermal conductivity of the pine is k_a and that of the brick is $k_d(= 5.0k_a)$. The face area A of the wall is unknown. Thermal conduction through the wall has reached the steady state; the only known interface temperatures are $T_1 = 25^\circ C$, $T_2 = 20^\circ C$ and $T_5 = -10^\circ C$. What is the interface temperature T_4 ?

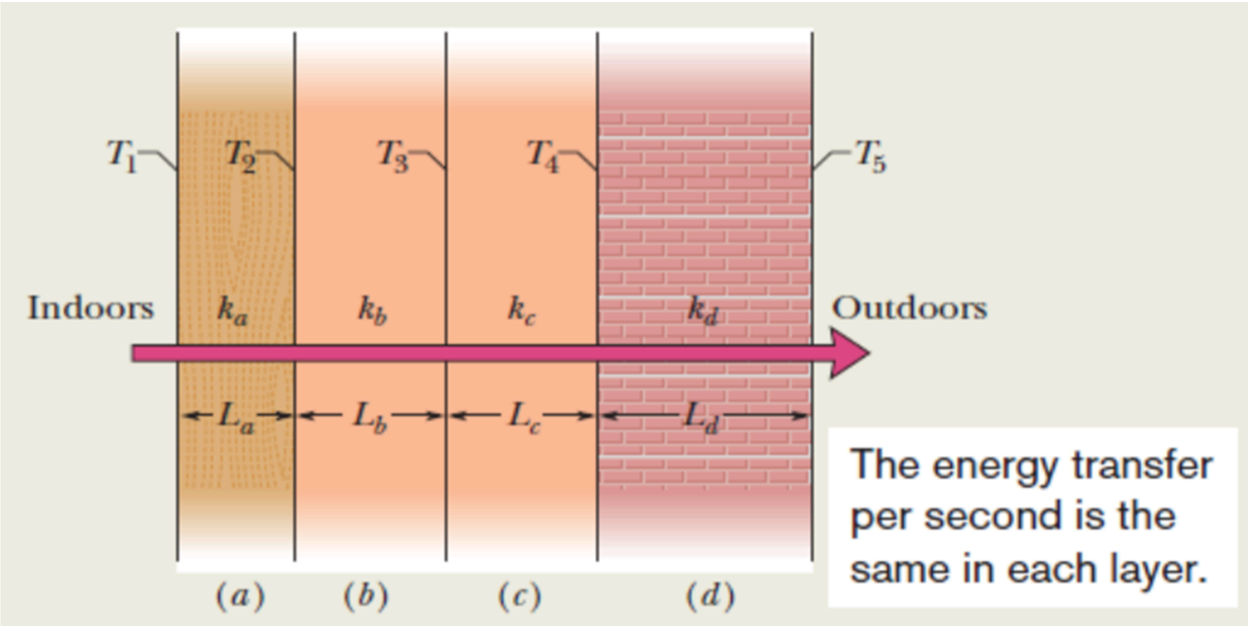
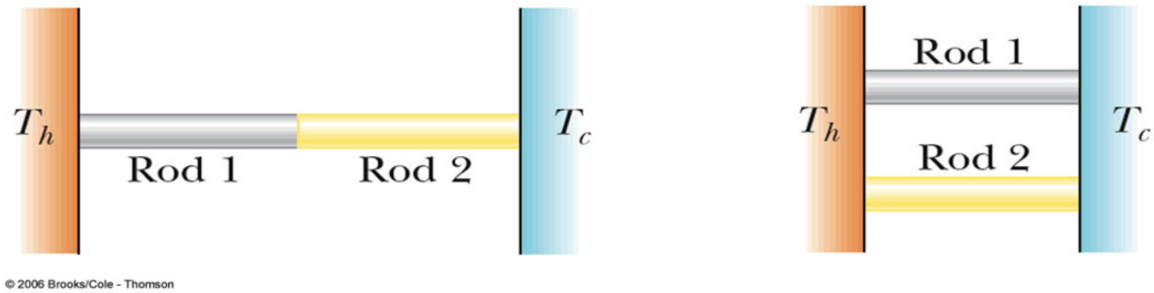


Figure 18-22 Steady-state heat transfer through a wall.

13. Two rods of the same length and diameter are made from different materials. The rods are connected between two hot and cold regions as shown (series and parallel). In which case is the heat transfer rate larger?
    1. Series 2. Parallel 3. Rate is same for both cases



14. An ideal absorber is defined as an object that absorbs all the energy incident on it. This type of object is called a ( ).

15. A cylinder contains 12 L of oxygen at 20°C and 15 atm. The temperature is raised to 35°C, and the volume is reduced to 8.5 L. What is the final pressure of the gas in

atmospheres? Assume that the gas is ideal.

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16. One container is filled with argon gas and another with helium gas. Both are at the same temperature. Which atoms have the higher rms speed?
    1. Argon 2. Helium 3. Same speed 4. Cannot tell

六、热力学进阶

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1. A bubble of 5.00 mol of helium is submerged at a certain depth in liquid water when the water (and thus the helium) undergoes a temperature increase ΔT of 20.0°C at constant pressure. As a result, the bubble expands. The helium is monatomic and ideal.
   (a) How much energy is added to the helium as heat during the increase and expansion?
   (b) What is the change in the internal energy of the helium during the temperature increase?
   (c) How much work W is done by the helium as it expands against the pressure of the surrounding water during the temperature increase?

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2. Initially, 1 mol of oxygen (assumed to be an ideal gas) has a temperature of 310 K and a volume of 12 L. We will allow it to expand to a volume of 19 L.
   (a) What would be the final temperature if the gas expands adiabatically? Oxygen (O_2) is diatomic and here has rotation but no oscillation.
   (b) What would be the final temperature and pressure if, instead, the gas expands freely to the new volume, from an initial pressure of 2.0 Pa?

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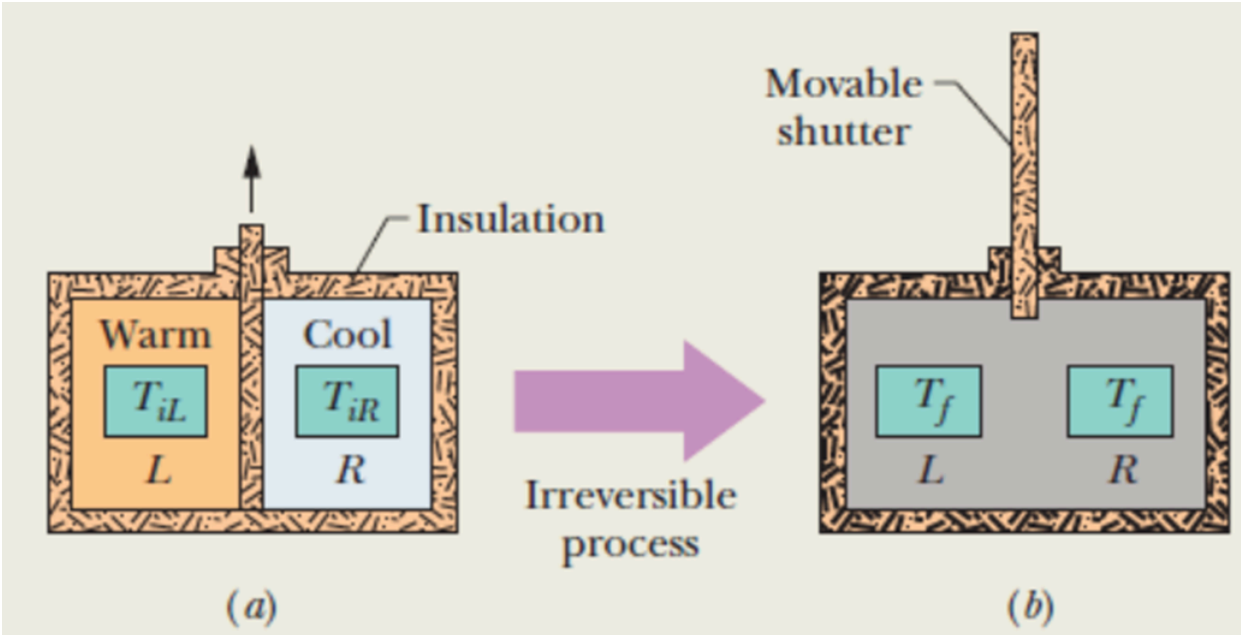
3. Figure shows two identical copper blocks of mass $m = 1.5\,\mathrm{kg}$: block L at temperature $T_{iL} = 60^{\circ}C$ and block R at temperature $T_{iR} = 20^{\circ}C$. The blocks are in a thermally insulated box and are separated by an insulating shutter. When we lift the shutter, the blocks eventually come to the equilibrium temperature $T_f = 40^{\circ}C$. What is the net entropy change of the two-block system during this irreversible process? The specific

heat of copper is 386 J/kg·K.



4. Suppose 1.0 mol of nitrogen gas is confined to the left side of the container. You open the stopcock, and the volume of the gas doubles. What is the entropy change of the gas for this irreversible process? Treat the gas as ideal.

5. What are the four processes of a Carnot Cycle?

6. How to determine the net work of a Carnot Cycle?

7. What is the difference between the Stirling engine and the Carnot engine? Please draw the P-V plot to show the above difference.

8. Imagine a Carnot engine that operates between the temperatures $T_H = 850\,\mathrm{K}$ and $T_L = 300\,\mathrm{K}$. The engine performs 1200 J of work each cycle, which takes 0.25 s.
(a) What is the efficiency of this engine?
(b) What is the average power of this engine?
(c) How much energy $|Q_H|$ is extracted as heat from the high-temperature reservoir every cycle?
(d) How much energy $|Q_L|$ is delivered as heat to the low-temperature reservoir every cycle?
(e) By how much does the entropy of the working substance change as a result of the energy transferred to it from the high-temperature reservoir? From it to the low-temperature reservoir?

9. A refrigerator is a device that uses ( ) in order to transfer energy from a low-temperature reservoir to a high-temperature reservoir.

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10. A refrigerator that did not require some input of work is a perfect refrigerator that transfers energy as heat Q from a cold reservoir to a warm reservoir without the need for work. How to verify that there does not exist the perfect refrigerator?

七、原子物理与半导体

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1. In order to explain the ( ) experiments, Rutherford proposed his atomic structure.

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2. What are the difficulties with the Rutherford Model of atomic structure?

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3. The Bohr Theory of Hydrogen includes both ( ) and non-classical ideas.

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4. In Bohr's Assumptions for Hydrogen, the electron moves in circular orbits around the proton under the influence of the ( ) force of attraction, which produces the centripetal acceleration.

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5. In order to explain the ( ) in Atomic Scattering Experiments, Sommerfeld improved Bohr's atomic structure.

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6. The Zeeman effect is the splitting of spectral lines in ( ).

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7. When a metal target is bombarded by high-energy ( ), x-rays are emitted.

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8. List out at least two substances emitted when a metal target is bombarded by X-ray. List out at least two substances emitted when a metal target is bombarded by high-energy electrons.

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9. Explain the mechanism of Characteristic X-Rays.

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10. An atom may have many possible energy levels. At ordinary temperatures, most of the atoms in a sample are in the ground state. Only photons with energies corresponding to ( ) can be absorbed.

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11. Once an atom is in an excited state, there is a constant probability that it will jump back to a lower state by emitting a ( ).

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12. ( ) is the prerequisite for laser generation.

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13. The ( ) and the ( ) population of the highest bands determines whether the solid is a conductor, an insulator, or a semiconductor.

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14. The valence band is the ( ) band. The ( ) band is the next higher empty band.

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15. Explain why conductors are easy to conduct electrons.

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16. Explain why insulators are difficult to conduct electrons.

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17. Why does the resistivity of semiconductors decrease with increases in temperature?

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18. What is an n-type Semiconductor?

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19. What is a p-type Semiconductor?

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20. What is the depletion region?

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21. The p-n junction has the ability to pass current in ( ) direction.